

Math 275: Calculus III

Instructor: Benjamin Ridenhour

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Office Hours: MWF 9:30–10:30pm (or by appt.)

Office: Brink Hall 327

Web: <https://bit.ly/2scoPgS>

Class Hours: MWF 1:30–2:20am

Class Room: TLC 022

Course Description

Math 275 is the third and final part of the standard three-semester calculus sequence. The distinct feature of this part of the course is its focus on the multi-dimensional analyses, as opposed to one-dimensional analyses that you learned in Math 170 (Calculus I) and Math 175 (Calculus II). This semester you will learn important concepts such as a vector, a vector field, a function of several variables, partial derivative, a line-integral and multi-variable integrals. You will see that these concepts—as scary as they may sound—are a natural generalization of the things you have previously learned. The ideas of the vector calculus apply to numerous areas of human knowledge such as engineering, physics, pure mathematics, biology, and many others. Some of the application will be seen in the course, some will surface later in your future courses, and others may not be seen until needed in the “real world.”

The structure of the class is as follows. Homework will be assigned for each class meeting. **All homework will be assigned and graded in the WebAssign platform, thus it is critical for you to sign up for an account immediately.** I expect you to have read the appropriate section materials and done the homework *prior to* each class period. Class periods will be used for minimal lecturing, discussion of the homework set, and strategies for solving problems. You should be prepared to participate in discussions. Student assessment is based around completion of the homework, *weekly quizzes on Friday*, 3 exams, and a final (comprehensive) exam (see below for details). Dates of the 3 exams will be determined based on progress of the class and announced at least a week in advance (they will roughly occur after every 4 weeks). I will drop your lowest quiz score.

You should be aware of several facets of the homework for the course. First, homework is worth a relatively small portion of your grade (only 10%). Second, each homework is equally weighted regardless of the number of questions in the homework. Third, for any question, WebAssign will give you *five* chances to answer the question. Fourth, you may request extensions for assignments which will be automatically granted; the only caveat is that questions/answers submitted after the due date will be worth 50% of the original value (note that correct answers

submitted before the deadline are worth 100% of the value, even if you do not complete the whole assignment). You can request extensions up to the end of the course (an extension grants a block of 14 days). In other words, you could not submit any homework until the last day of class and—assuming you got all of the questions correct—still manage to get 5% of the available 10% for doing homework assignments. **Aside from the actual assignment given on WebAssign/Canvas, I strongly recommend you do as many of the problems at the end of each section as possible for practice; answers are provided for all questions in the e-book.**

Scheduling oddity: Due to Jazz Fest, TLC 022 will not be available 16 and 18 April; we will not have class those days.

Required Materials

- *Calculus*. 9th edn. by J Stewart, D. Clegg, and S. Watson. Cengage Learning: Boston.
- **Subscription to Cengage WebAssign** (*Note:* Access to WebAssign is part of Inclusive Access along with the e-book. Otherwise, the bookstore can sell access or you can pay for access directly via the WebAssign website.)

Prerequisites/Corequisites

Prerequisites: Math 175

Topics

1. Vectors and Geometry of Space (Chapter 12)
2. Vector Functions (Chapter 13)
3. Partial Derivatives (Chapter 14)
4. Multiple Integrals (Chapter 15)
5. Vector Calculus (Chapter 16)

Course Objectives/Learning Outcomes

Successful students will:

1. Develop the ability to reason geometrically in more than two dimensions, including mastery of basic notions of vector geometry and equations of lines, planes, and spheres.
2. Be able to think geometrically about the graphs of functions of two variables, including the significance of partial derivatives, directional derivatives, and gradients.
3. Learn to set up and evaluate integrals in two and three variables, including working with any of the standard coordinate systems.

4. Understand and correctly interpret the basic notions of vector calculus, including line integrals of vector fields and Green's Theorem.

Final Exam Date

5 May (Monday), 12:45–2:45pm

Grading Policy

A typical grading scale will be used (i.e., $\geq 93\% = A$, $(93,90\%] = A-$, $(90,87\%] = B+$, ...). I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments using the following proportions:

- 10% of your grade will be determined by completing homework.
- 25% of your grade will be determined by weekly quizzes.
- 45% of your grade will be determined by 3 equally-weighted exams.
- 20% of your grade will be determined by the final exam.